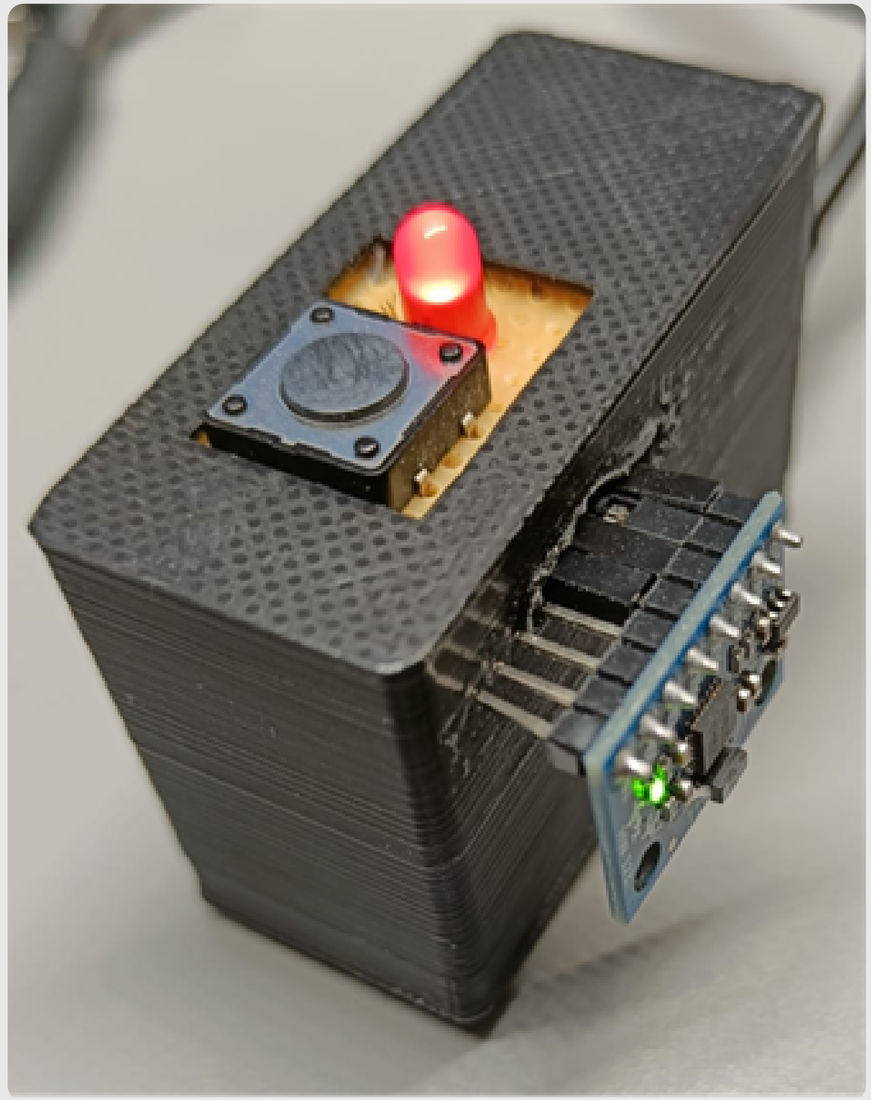
SMARTGUARD
A SMART WEARABLE FOR IMPACT AND FALL DETECTION



COURSE
INS4000 – Innovation with sensor technology
A.A. 2025/2026

GROUP MEMBERS
Marco Chimienti
Marcos Moura
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PROBLEM AND **MOTIVATION**

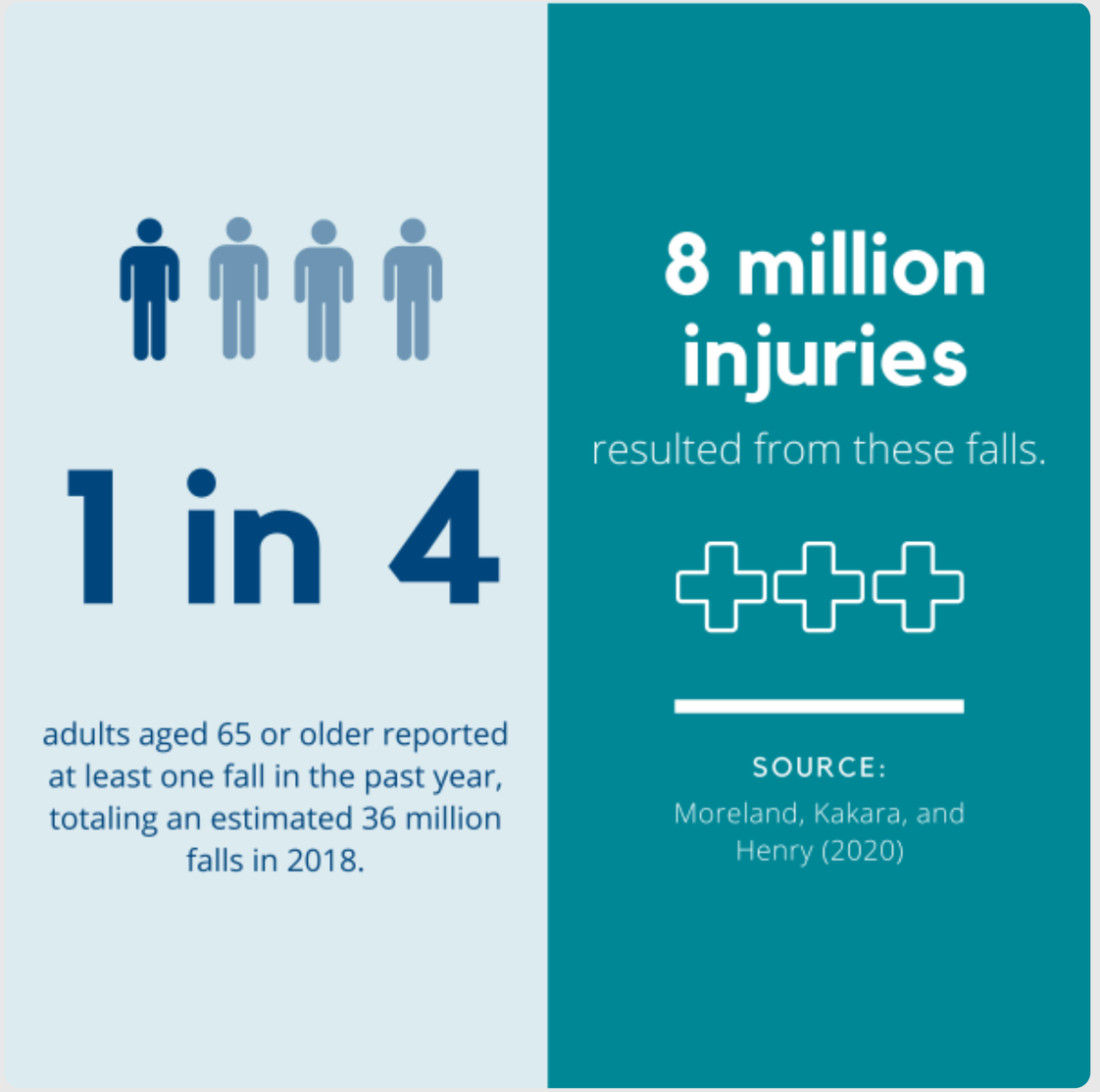
WHY DO WE NEED FALL DETECTION SYSTEMS?

Falls are a major cause of injury, hospitalizations and reduced quality of life, especially for elderly people and construction workers.

There is a need for a integrated, reliable and easy to use device.

EXISTING SOLUTIONS

- Vision-based → need cameras and infrastructure.
- Environment sensors → limited to a room.
- Smart wearables → heavier, more power consumption.



RECENT LITERATURE

RECENT LITERATURE OF SMART WEARABLE DEVICES (2021 – 2025)

SENSORS AND PLACEMENT

- Most prototypes use 3-axis accelerometer, and a gyroscope.
- Some works add barometers, pressure insoles, physiological sensors (T, SpO2, gas).
- First, head, torso or waist.

DETECTION ALGORITHM

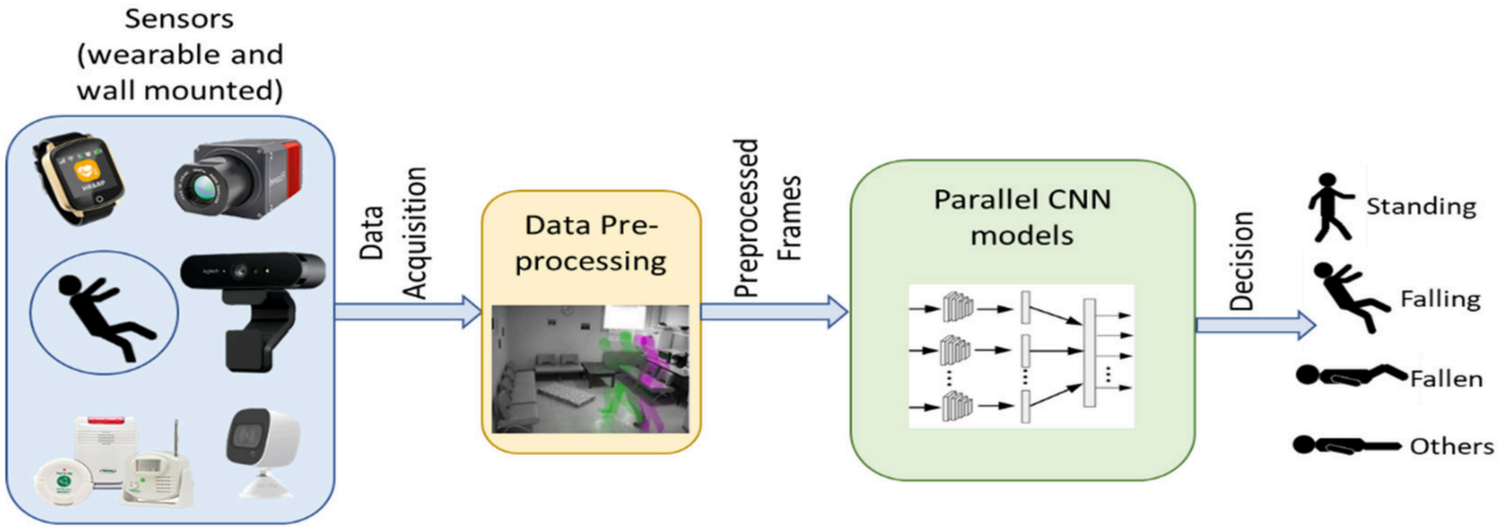
- Threshold-based, machine leaning models (CNN, DNN) and hybrid approaches.
- Threshold use simple pattern, sensitive to placement.
- Deep and hybrid models reach higher accuracy, but may need cloud computing.

TRANSMISSION METHOD

- Short range radios (Wi-Fi, Bluethooth and ZigBee) sending data to a smarthphone or gateway.
- Long-range, low-power such as LoRa and other LPWANs for remoting.
- Microcontroller + radio impact autonomy. Better separate modules.

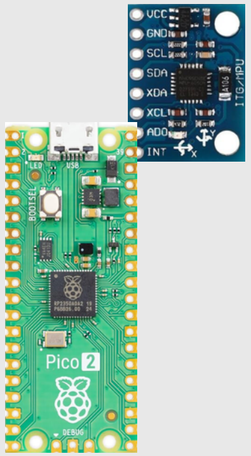
PRECISION AND FALSE ALARMS

- Thresold systems report around 85 – 90%; hybrid and deeep leaning 95 – 98% accuracy.
- Flase alarms caused by daily activities and sub-optimal sensor locations.
- Studies evaluated on small groups of young, healthy: robustness not well defined for elderly.

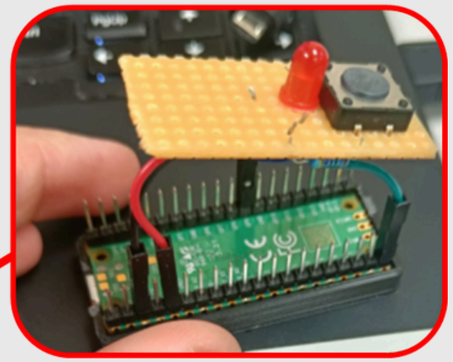


OUR SOLUTION

OBJECTIVES OF THE SMARTGUARD PROTOTYPE







- Use a **single IMU** (MP-6050) and a Raspberry Pi Pico as the processing unit
- Design and implement a **belt-worn** wearable device.
- Provide local feedback through a **LED** and a **button**, and real time data visualization through an interface .
- Develop a reliable **threshold-based** fall detection algorithm.

SYSTEM ARCHITECTURE

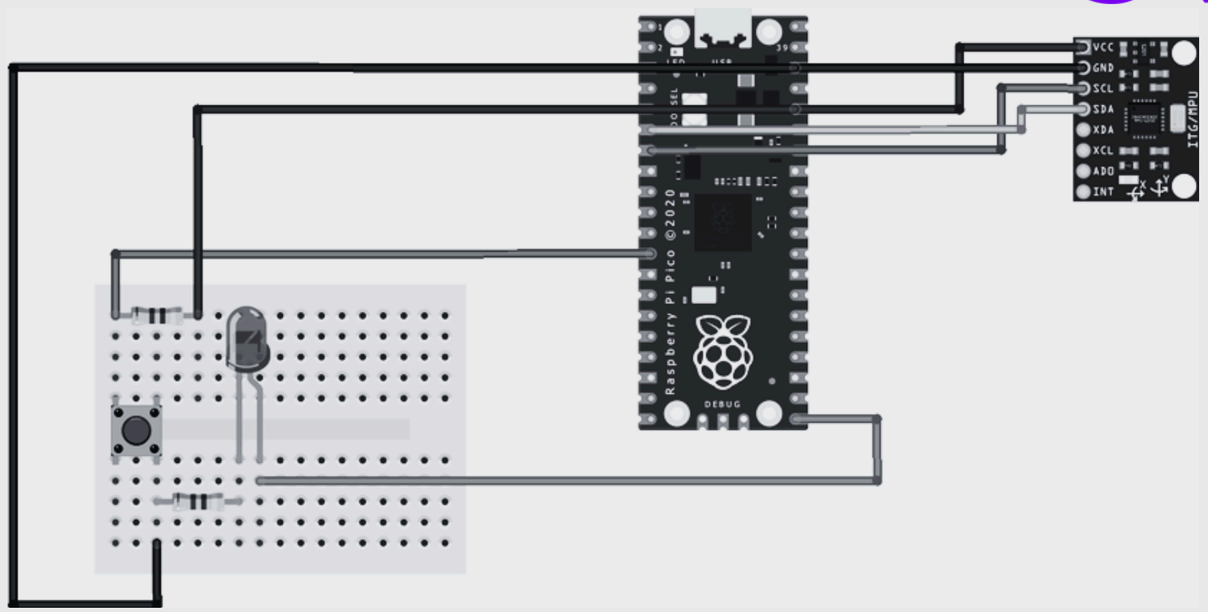
OVERALL SYSTEM LAYERS ARCHITECTURE

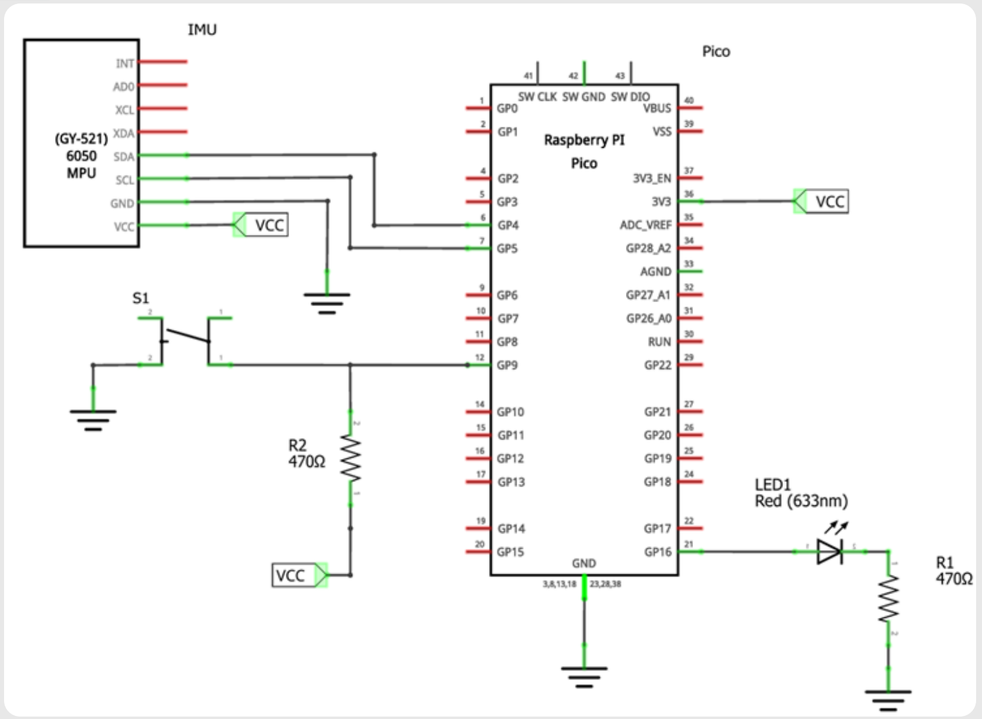
SENSING LAYER → MPU-6050 IMU measuring 3-axis acceleration, tilt and roll.

PROCESSING LAYER → Raspberry Pi Pico running the fall detection algorithm in C.

DECISION LAYER → Detection of fall events based on filtered signals and thresold

OUTPUT LAYER → LED and button on the device; Communication to a user interface for monitoring





SENSING & **PLACEMENT**

OVERVIEW AND BODY PLACEMENT

BODY PLACEMENT

- Device is worn on the waist, close to the body's center of mass.
- 3D-printed cover designed to be worn on a belt, or attached to a pocket

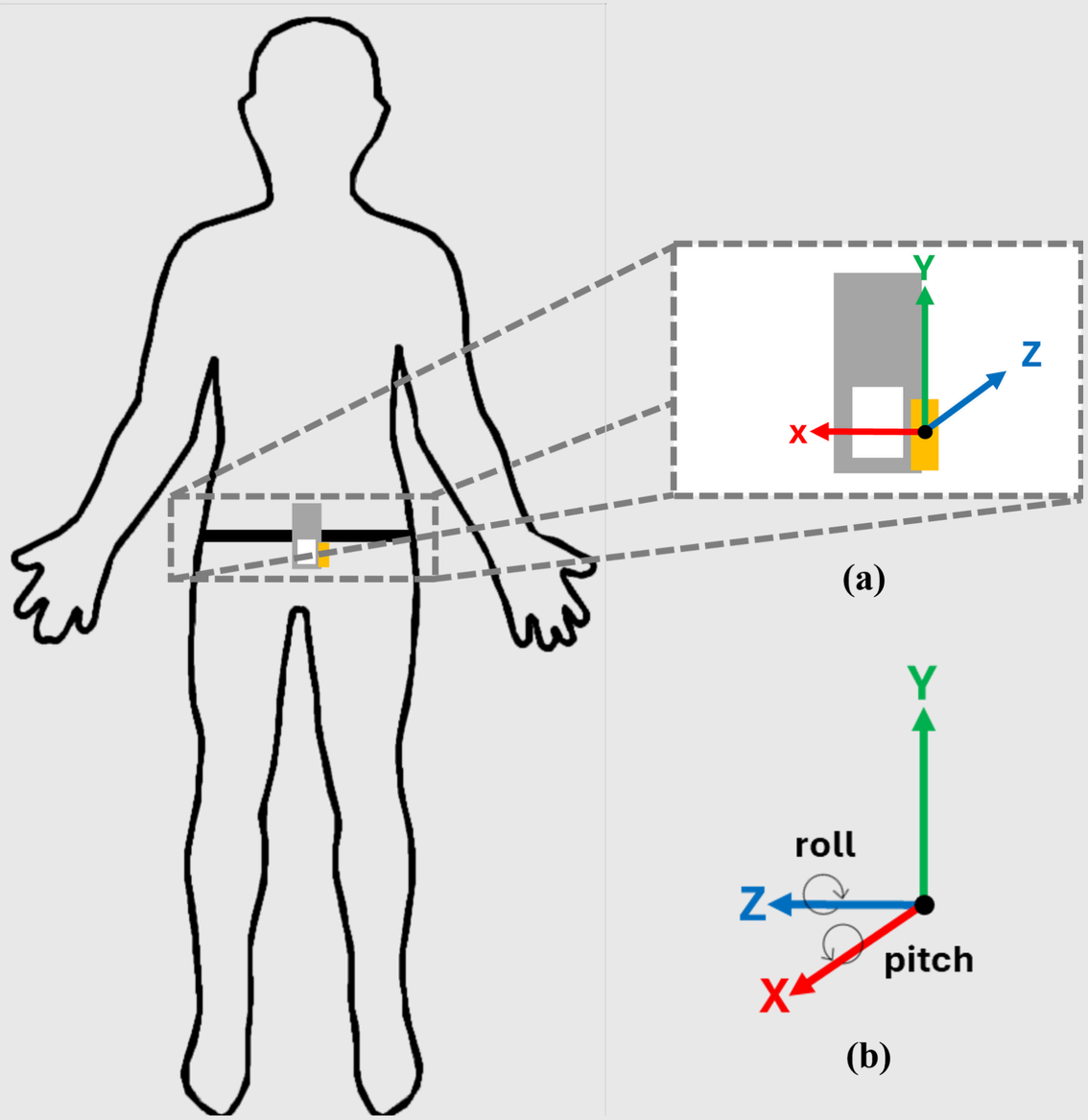
COMPUTED VARIABLES

- From accelerometer and gyroscope readings we compute:

1. Tilt angles around X and Z axes (pitch and roll)
2. Acceleration magnitude.

SENSORS

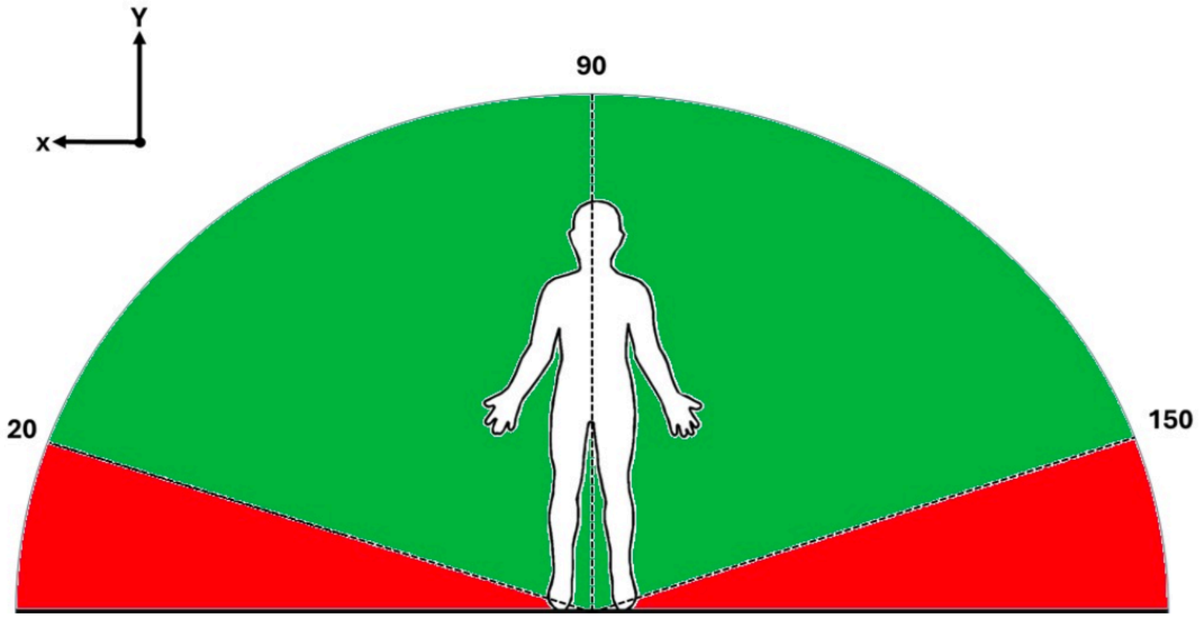
- IMU: MPU-6050 (accelerometer + gyroscope)
- Y-axis of the IMU aligned with the body axis



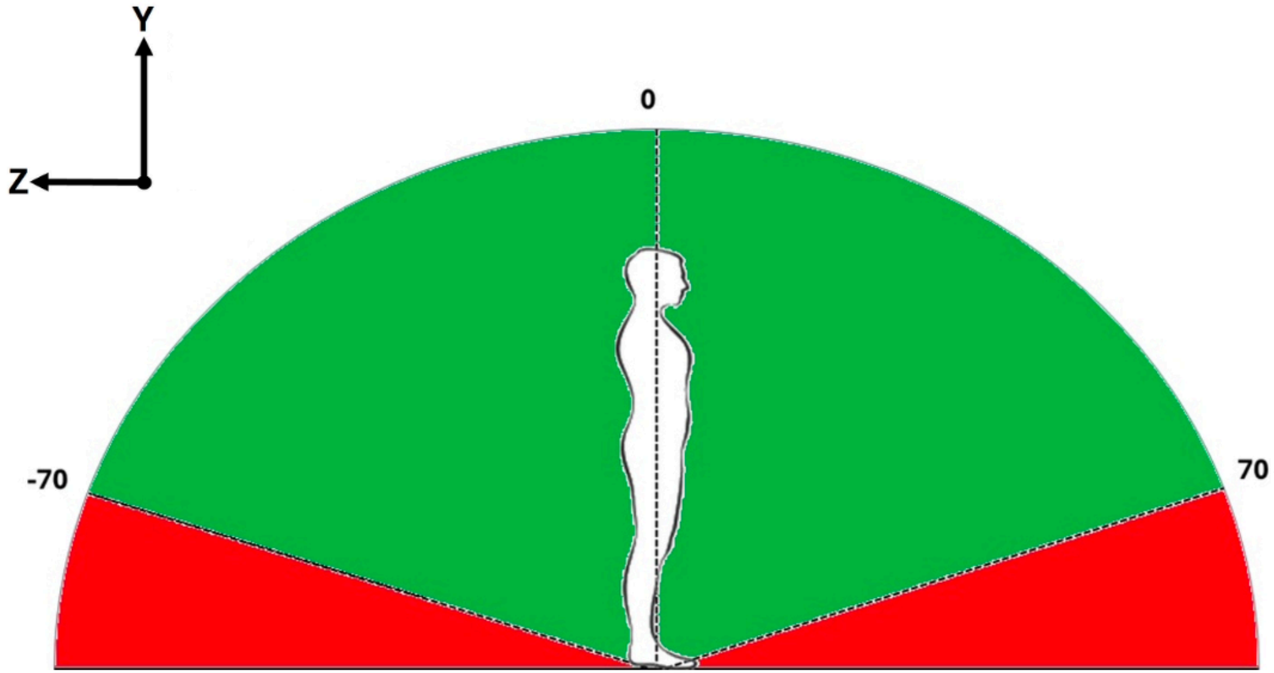
SENSING & **PLACEMENT**

PITCH AND ROLL

ROLL ANGLE RANGE



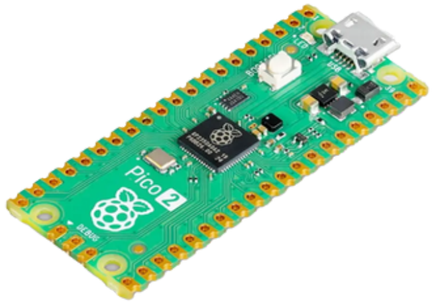
PITCH ANGLE RANGE



HARDWARE: **CIRCUIT, PCB**

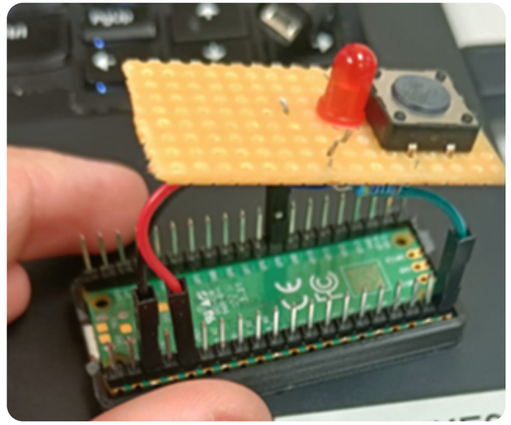
HARDWARE: FROM BREADBOARD TO WEARABLE DEVICE





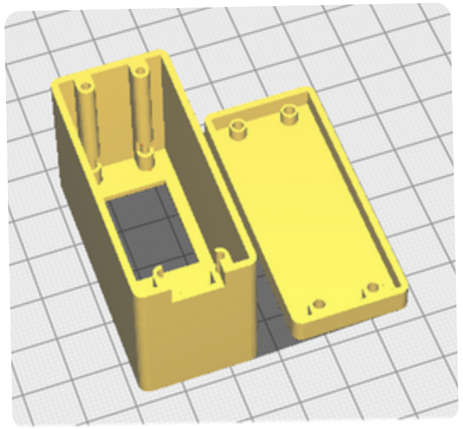
MICROCONTROLLER

- Microcontroller: Raspberry Pi Pico more powerful and smaller than a Arduino Uno board, and extremely cheap, less than 50 NOK



IMU CONNECTIONS

- IMU connected via I^2C, LED and button connected to GPIO pins



WEARABLE PACKAGE

- The wearable package was printed with a 3D printer. The original design had flaws that were manually corrected

FALL DETECTION ALGORITHM **OVERVIEW**

- We implemented a threshold-based algorithm to run directly on the microcontroller.
- The algorithm detects a fall as a sequence of kinematic events, instead of a single condition.

→

Main stages:

1. Free-fall: low acceleration magnitude (close to 0 g)
2. Impact: strong peak in acceleration magnitude.
3. Posture change: transition from standing to lying position using pitch and roll.
4. Post-fall conditions: event counter for entering the last step.
5. Confirm-fall check: if lying and still, alarm get triggered. LED on with a button.

WHAT'S INNOVATIVE?

Our algorithm implement temporal logic, posture information and different conditions in the datas acquired:

- Multi-stage reasoning: the decision is taken considering samples in a time window, making sure to conduct a proper analysis.
- The thresholds are combined with analysis of pitch and roll, which gives crucial information about the posture.
- Before entering the alarm, a post-fall condition check if at least two of the three conditions have been detected, activating the confirm-fall check.

ALGORITHM:
PRE-PROCESSING

DATA ACQUISITION, CALIBRATION AND PRE-PROCESSING

↗

STEP 1: DATA ACQUISITION AND CALIBRATION

- Reading data from the IMU in 50 Hz frequency.
- Static calibration, measuring static errore at rest and subtracting offsets to each sample. Removed bias, making pitch/roll estimation more accurate.

↓

- Accelerometer: -0.3 instead of 0 in X; -0.2 instead of 0 in z; 9.01 instead of 0.81 in Y.

$$a_z = a_{z,raw} + 0.2 \qquad a_x = a_{x,raw} + 0.3 \qquad a_y = a_{y,raw} + 0.9$$

- Gyroscope measured -3.5 °/s on X and -1.5 °/s on Z at rest. Added +3.5 and +1.5..

STEP 2: PREPROCESSING

- Low-pass filtering of the acceleration magnitude to reduce noise.
- Kalman filter applied to pitch and roll for more stable orientation estimates.

ALGORITHM: PROCESSING

THRESHOLD, MACHINE LEARNING

STEP 3: TRESHOLD

1. Free fall when acceleration magnitude: $mag_g < 0.4g$

2. Impact when: $mag_g > 2.5g$ within $t - t_{free} \geqslant 200ms$

3. Posture change when change in Z and X acceleration within: $(t - t_{tilt} \leqslant 500ms)$

$$|a_z| < 0.5 \,||| a_x| > 0.5$$

4. Post-fall monitoring: user remains almost still in a lying posture.

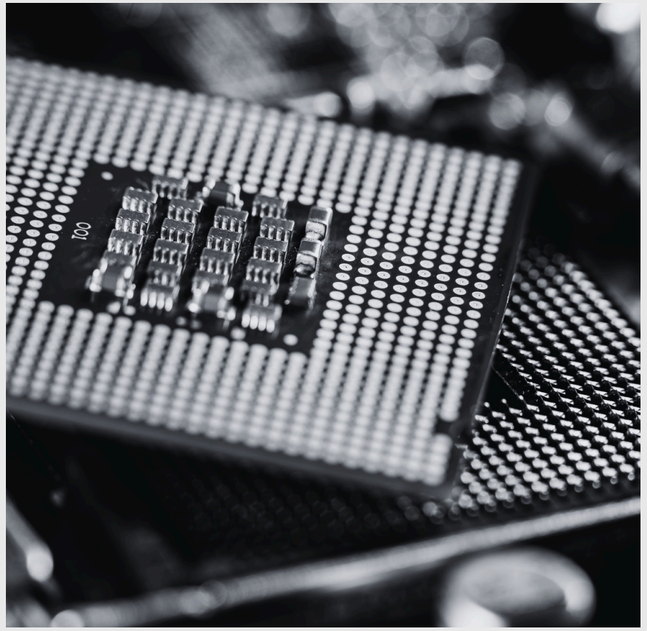
$$eventCount = (FreeFallEvent) + (ImpactEvent) + (TiltChangeEvent \,||isLyingNow)$$

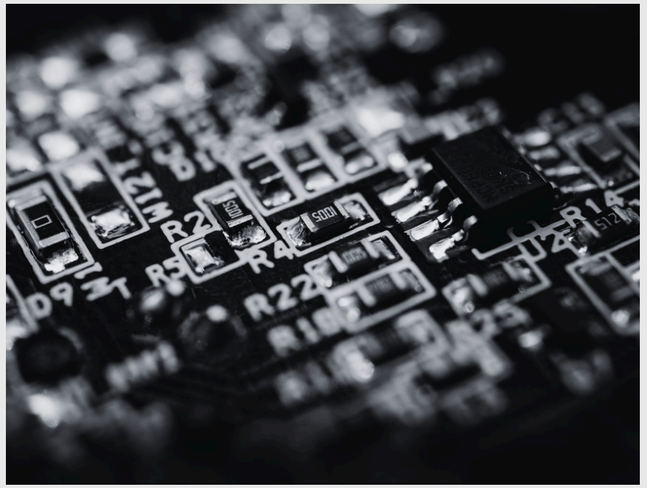
5. Post-fall check trigger the alarm when, within 3 seconds

$$isLyingNow_{post} = (|roll| < 20^{\circ} \&\& \,(|roll| > 150^{\circ} \,|||itch| > 70^{\circ}) \qquad 0.75 \leqslant mag_g\,(t) \leqslant 1.3$$

STEP 4: MACHINE LEARNING?

- A CNN-BiLSTM model trained offline on the MobiFall dataset to classify time windows as fall, or as activity of daily life.
- Accuracy of 96%. Next prototype as hybrid solution: thresholds for fast on-device filtering, ML for refined classification.

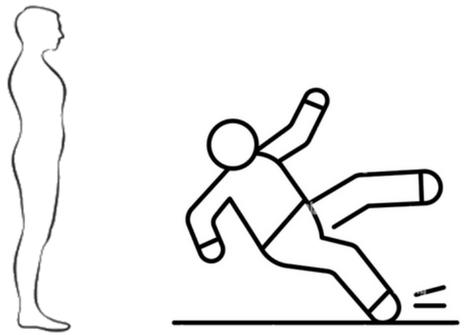




EXPERIMENTAL **SETUP**

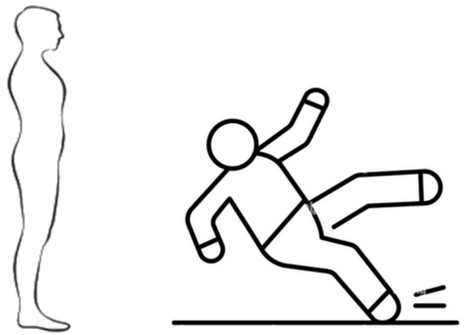
STATIC TESTS

(1) (2)



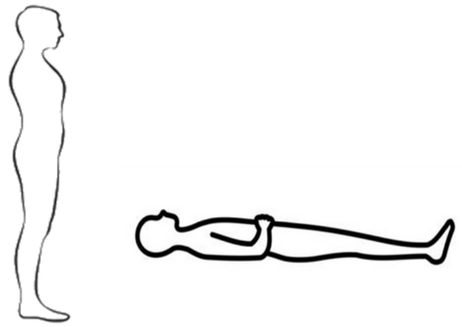
LATERAL FALL (Z-AXIS)

(1) (2)



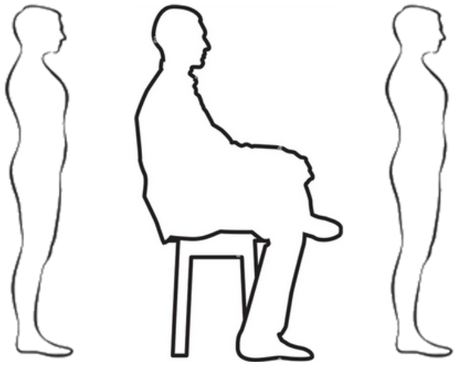
FRONTAL FALL (X-AXIS)

(1) (2)

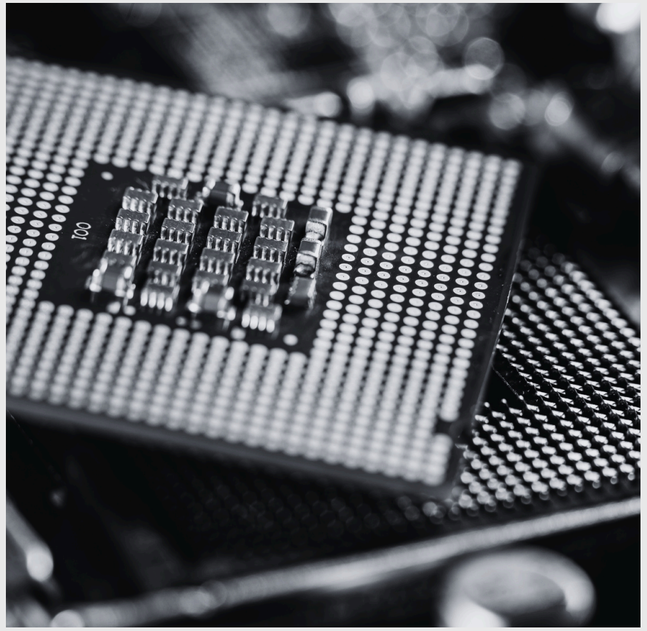


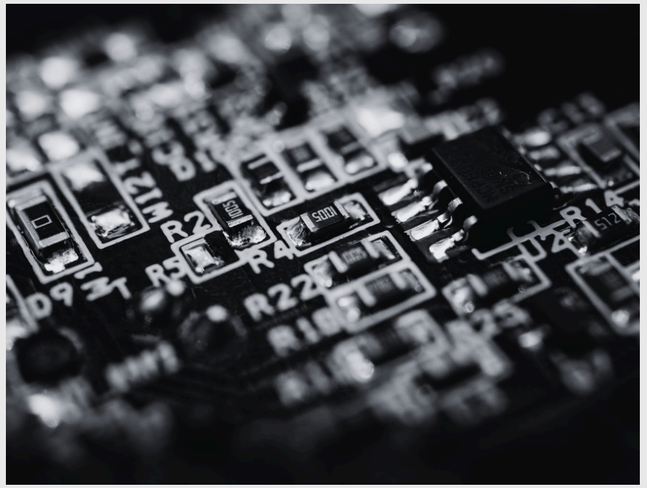
LYING DOWN

(1) (2) (3)



SITTING / STANDING

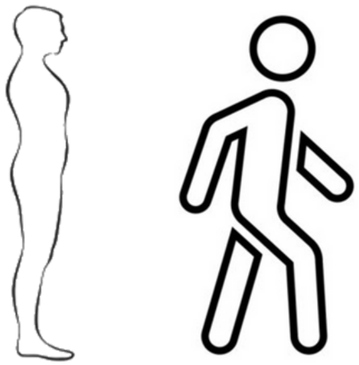




EXPERIMENTAL **SETUP**

DYNAMIC TESTS

(1) (2)



WALKING



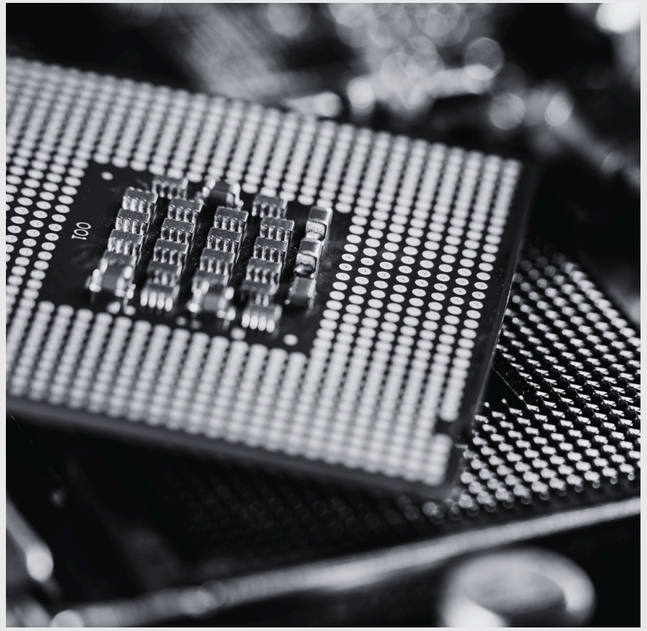
PLAYING WITH DOG

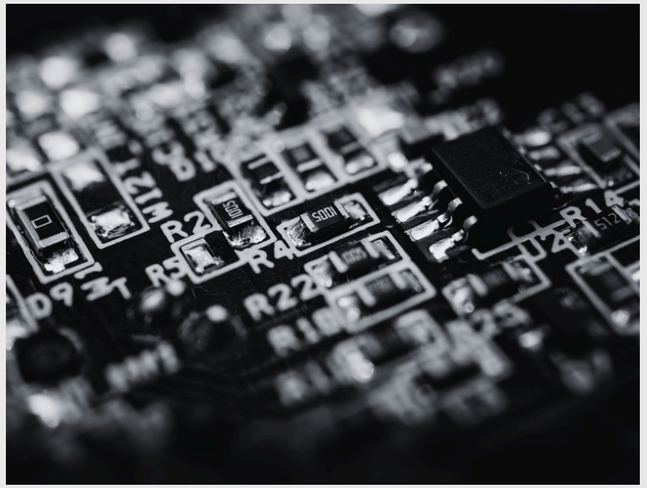


RUNNING



JUMPING





EXPERIMENTAL **SETUP**

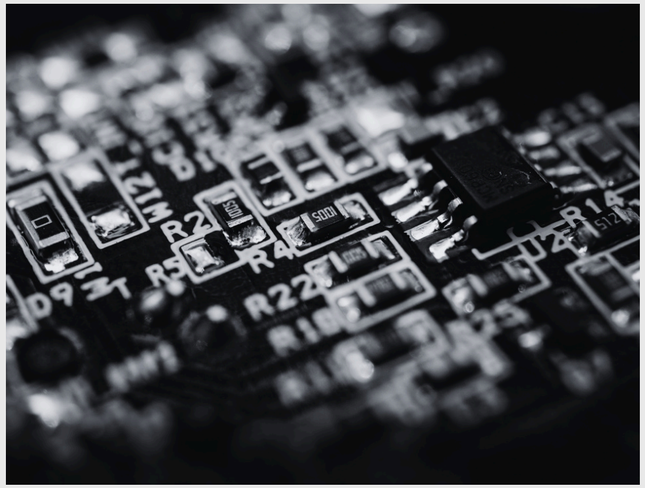
PERFORMANCE EVALUATION

- Simulate 20 falls
  - Get average time of fall detection
  - Get accuracy
- Compare the time with Threshold-based studies
  - [1] D. B. Apatsara Kitkamchon, Kan Dissorn, “A wearable fall detection device,” The 2021 Biomedical Engineering International Conference, vol. 21, no. 6, 2021
  - [2] M. Villa and E. Casilari, “Energy-efficient fall-detection system using lora and hybrid algorithms,” Biomimetics, vol. 10, no. 5, p. 313, 2025.
  - [3] J. Fernandez-Bermejo, J. Martinez-del Rincon, J. Dorado, X. D. Toro, M. J. Santofimia, and J. C. Lopez, “Edge computing transformers for fall detection in older adults,” International journal of neural systems, vol. 34, no. 05, p. 2450026, 2024.

(1) (2)

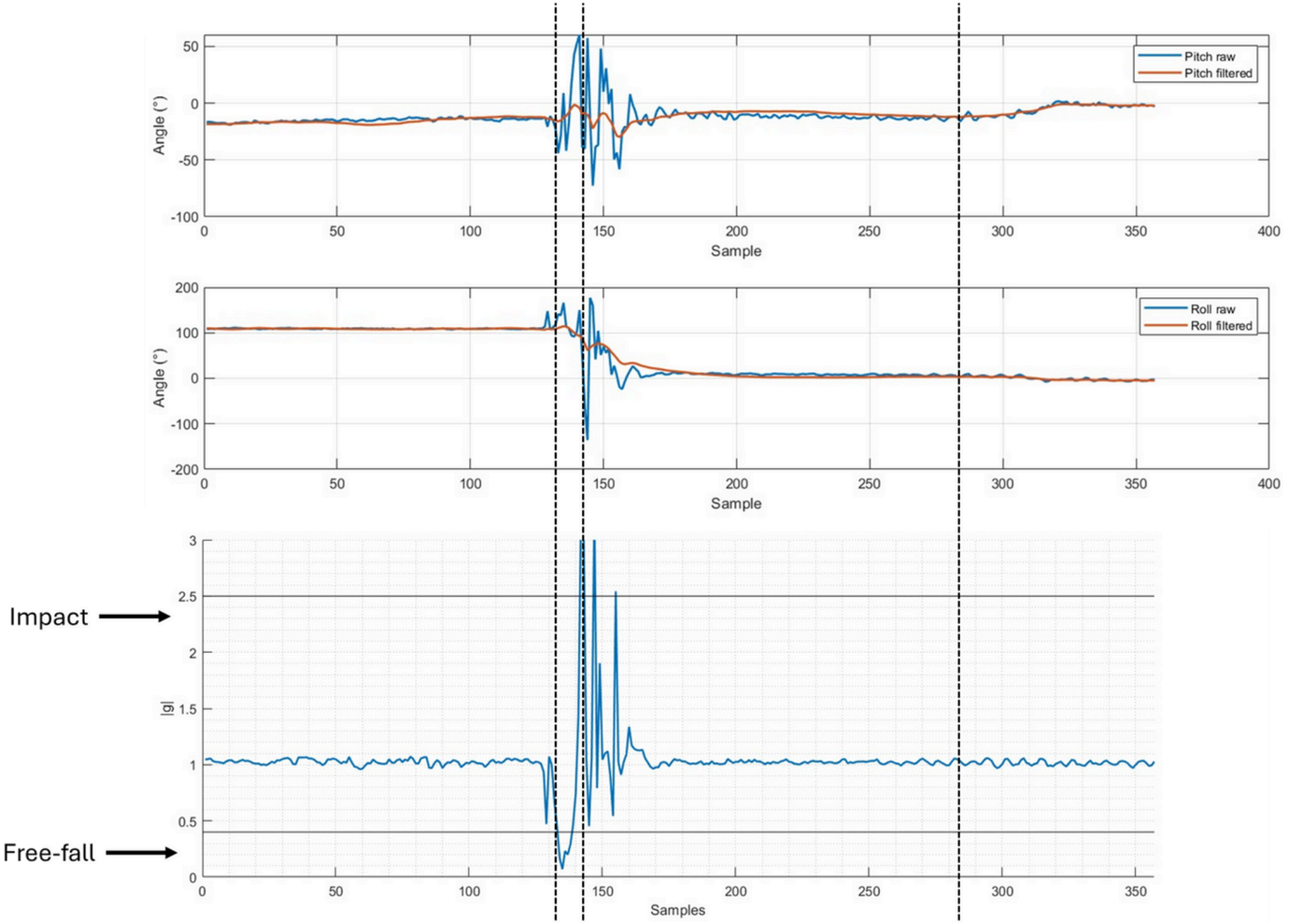






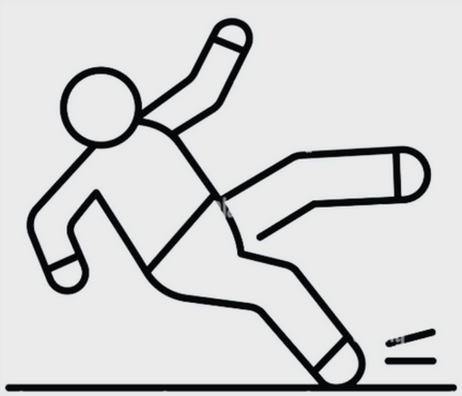
RESULTS

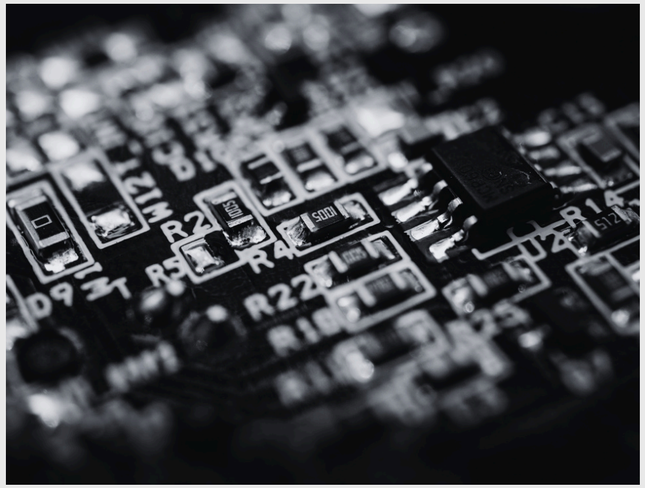
LATERAL FALL (Z-AXIS)



(1) (2)

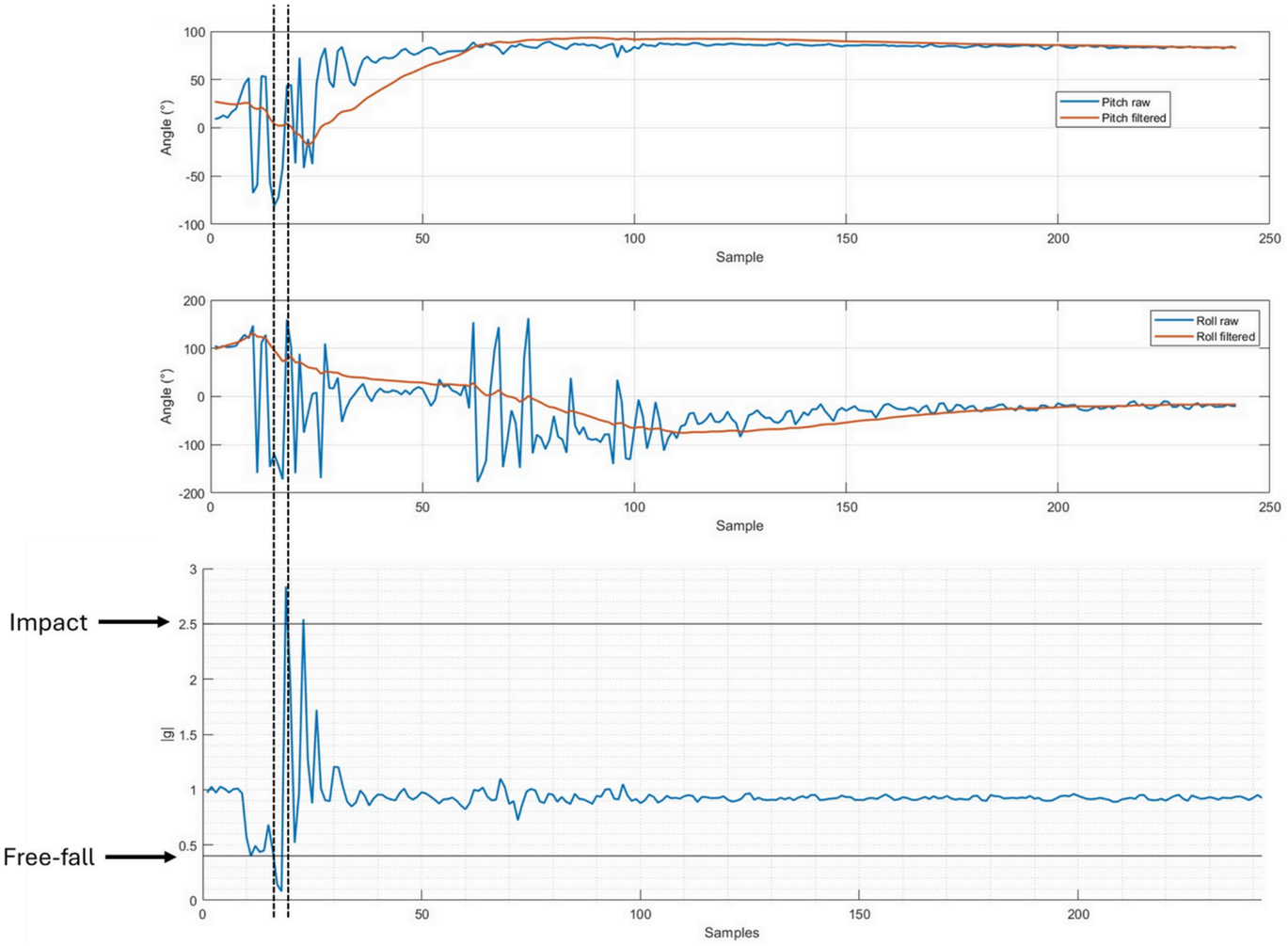






RESULTS

FRONTAL FALL (X-AXIS)

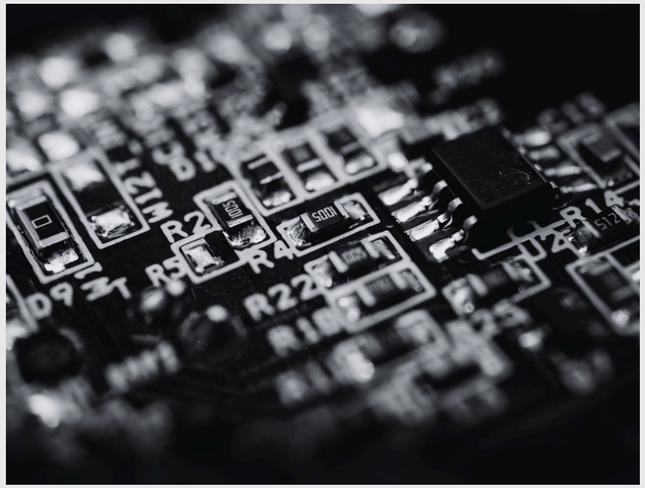


RESULTS

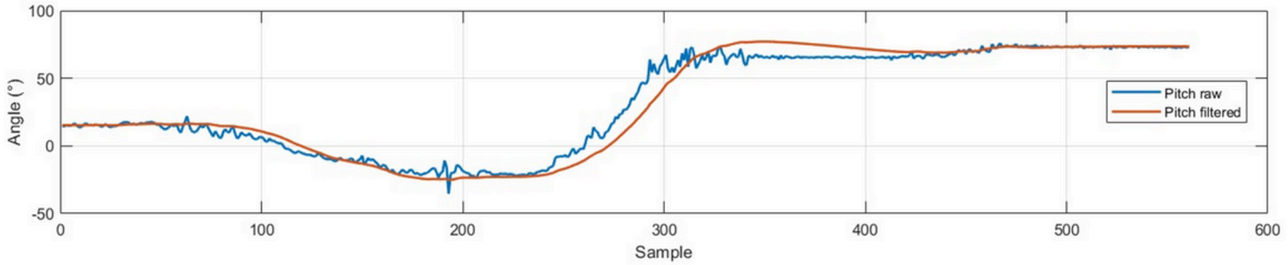
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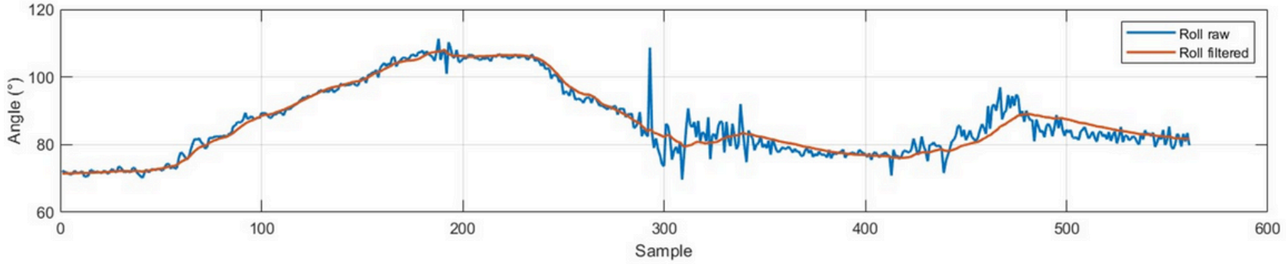


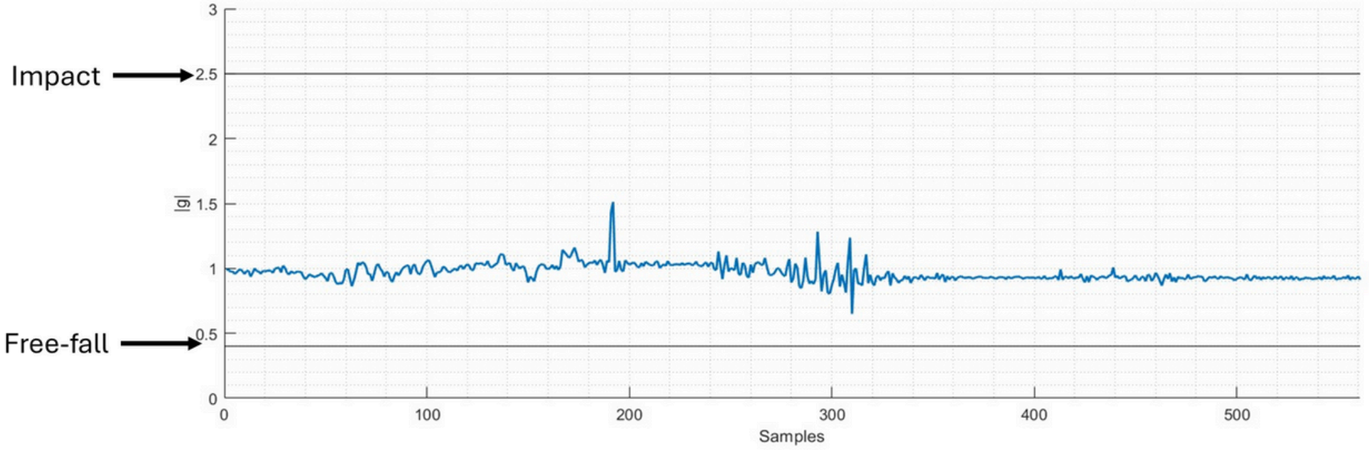


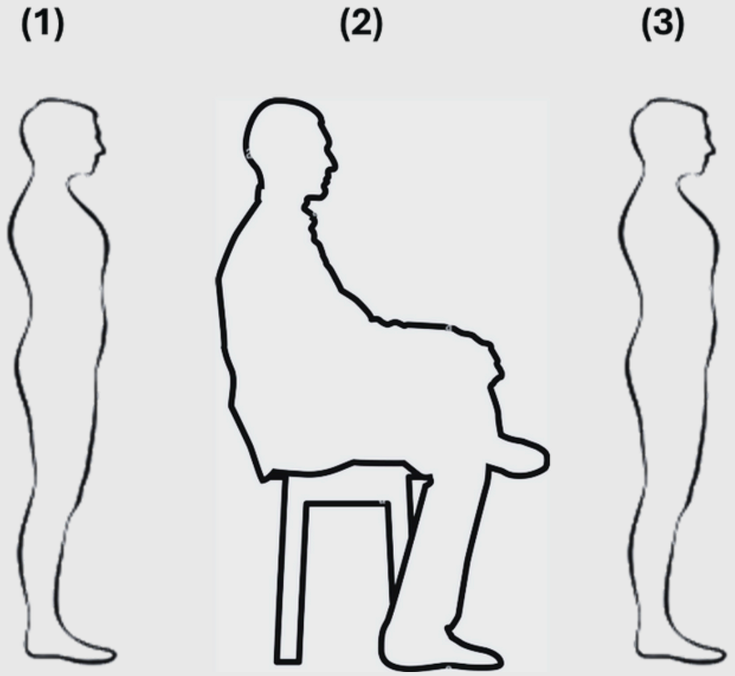


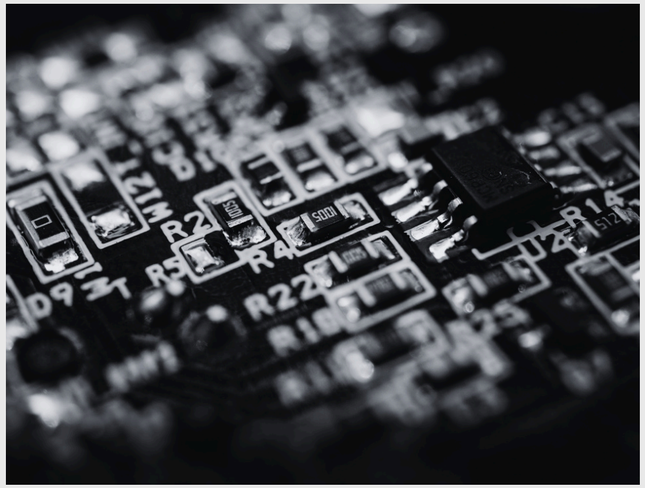
LYING DOWN





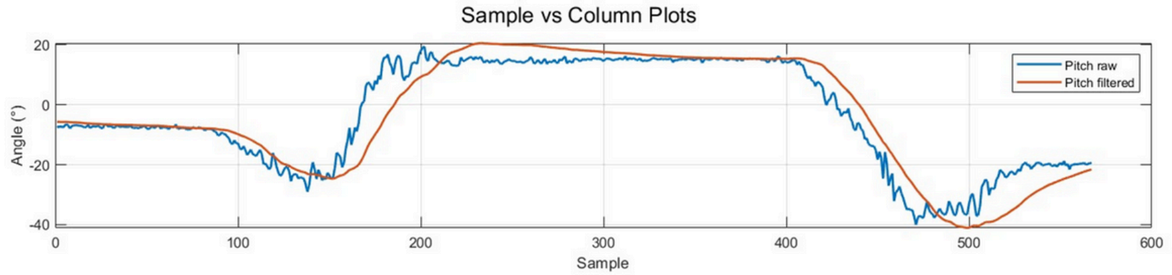


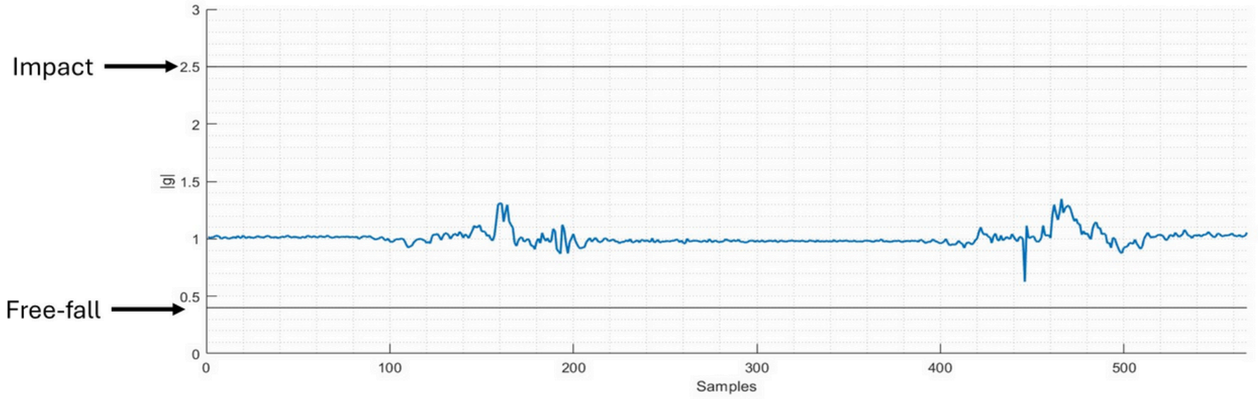




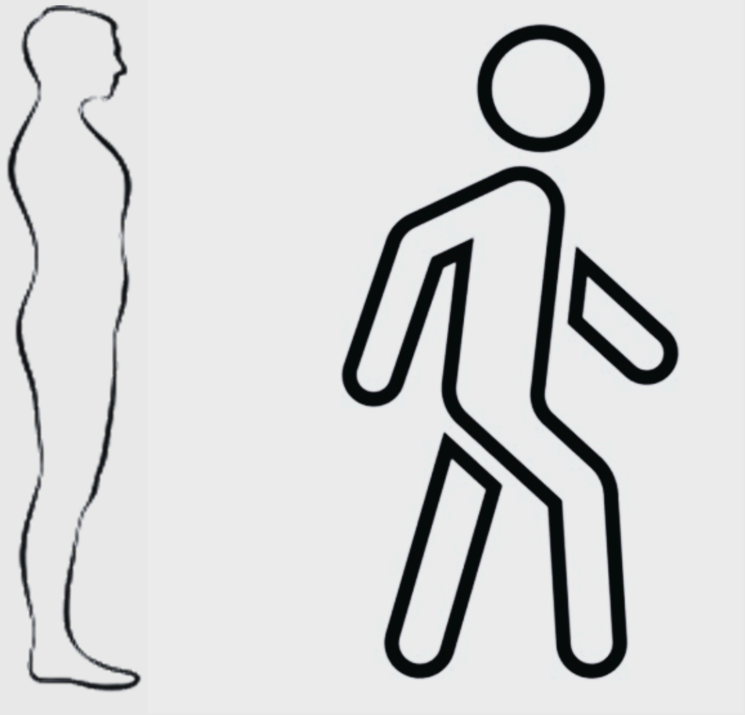
RESULTS

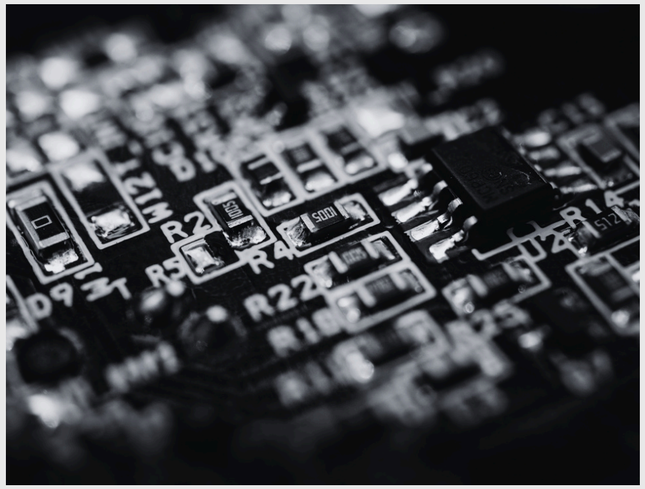
SITTING/STANDING





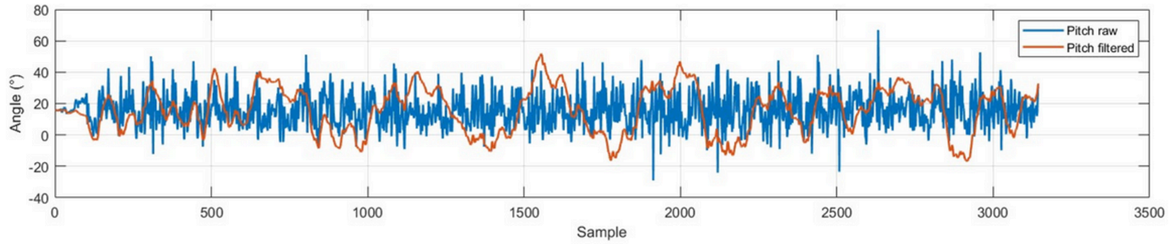
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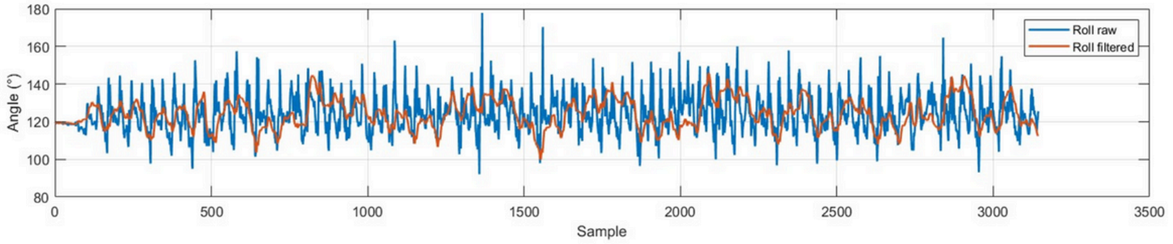


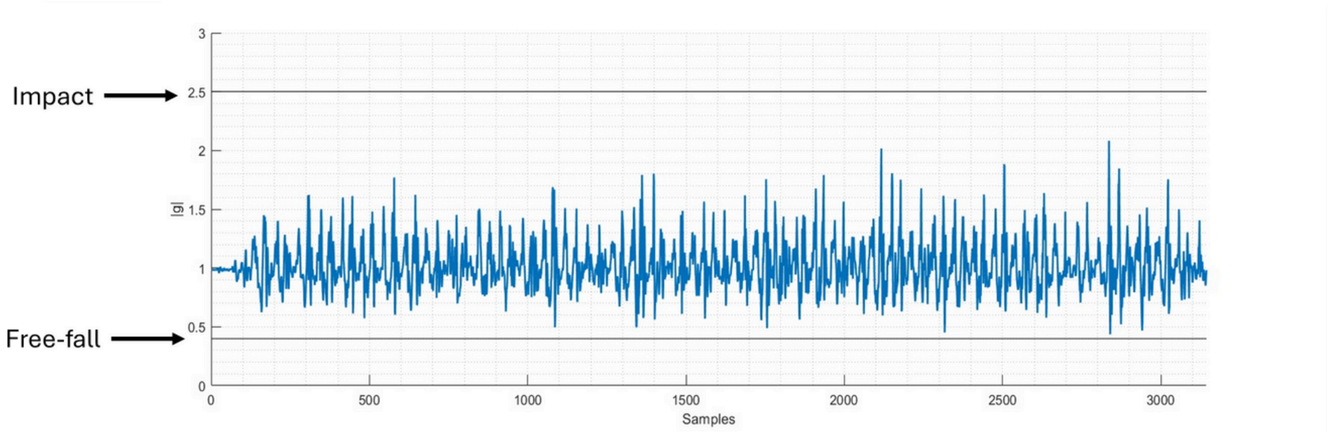


RESULTS

WALKING





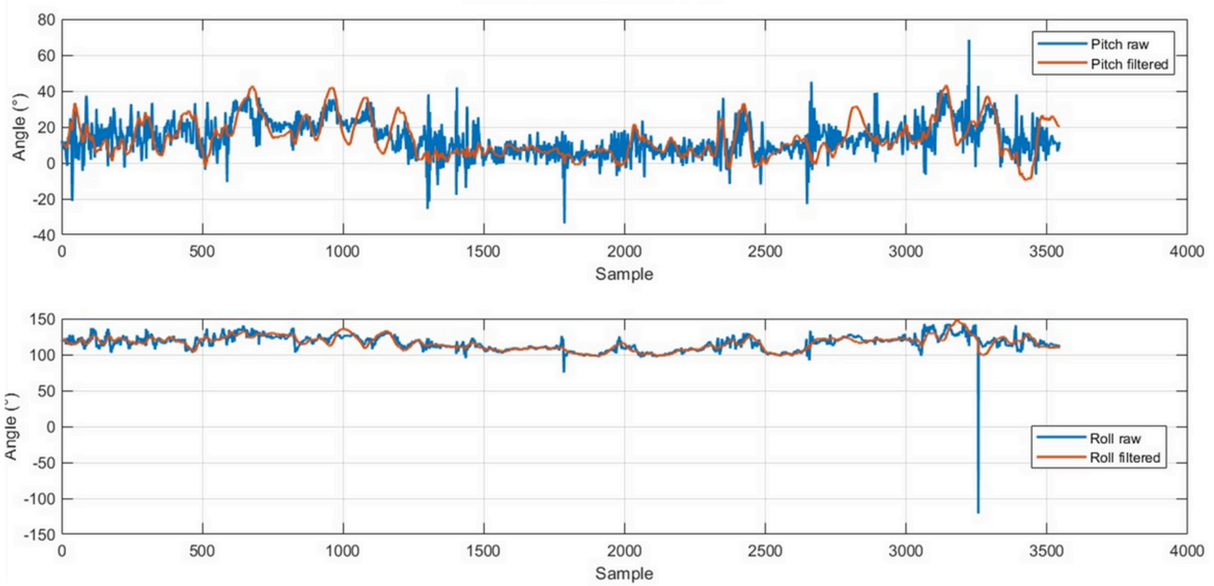


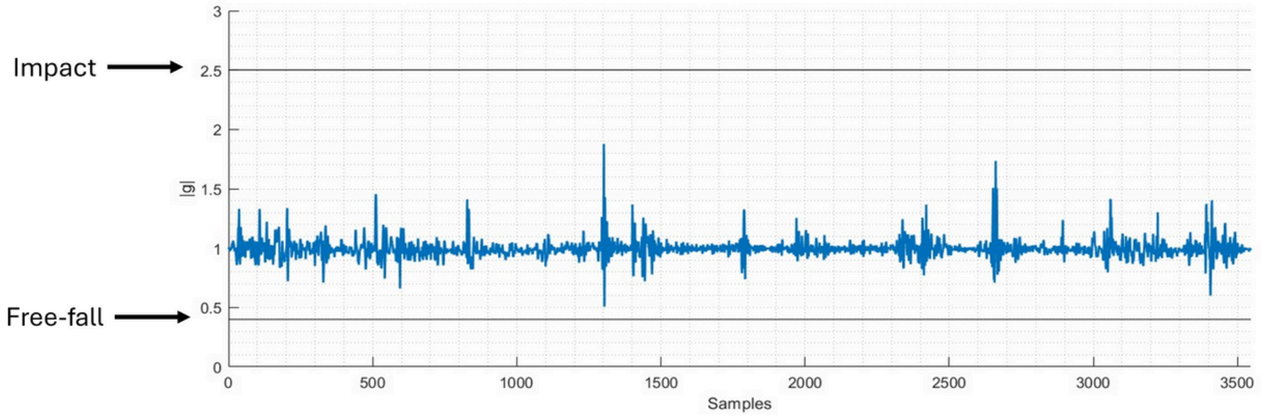




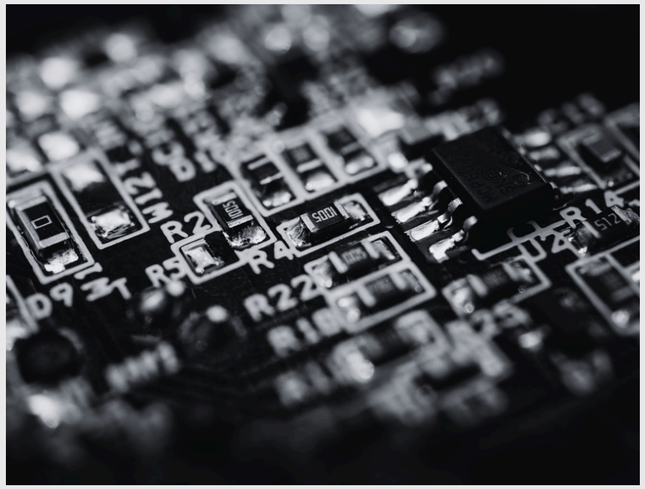
RESULTS

PLAYING WITH DOG



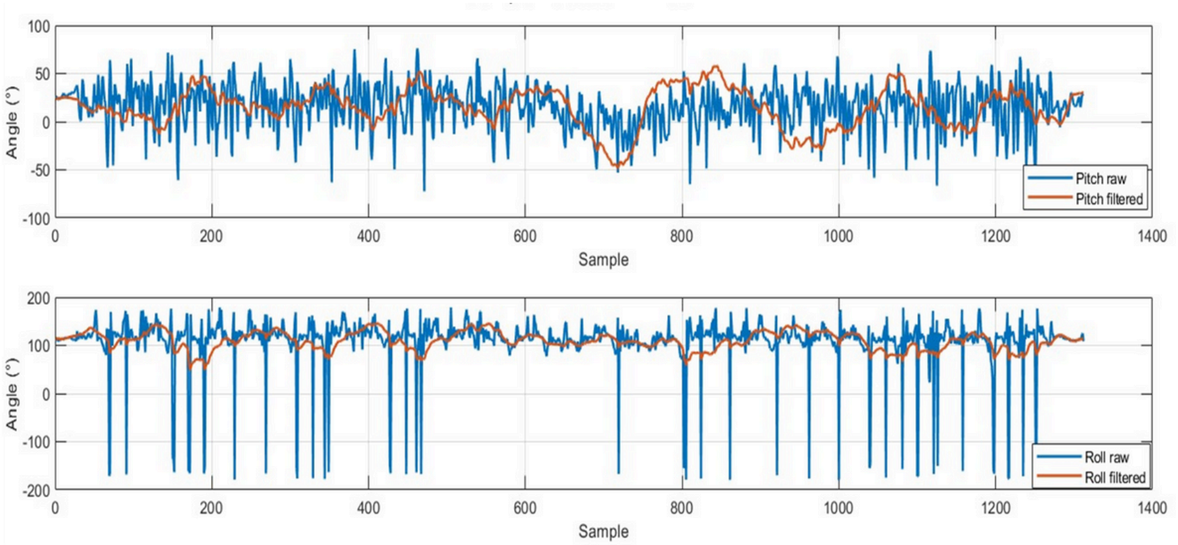


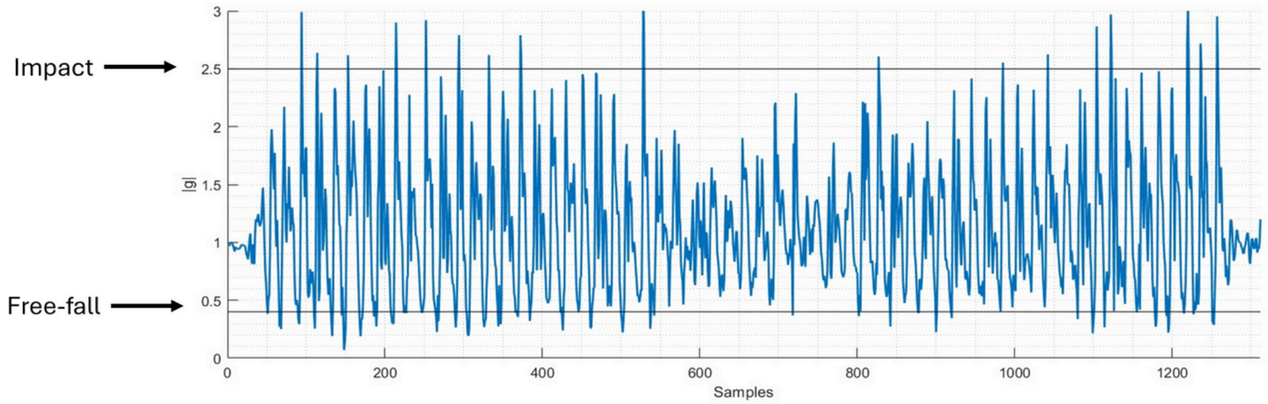




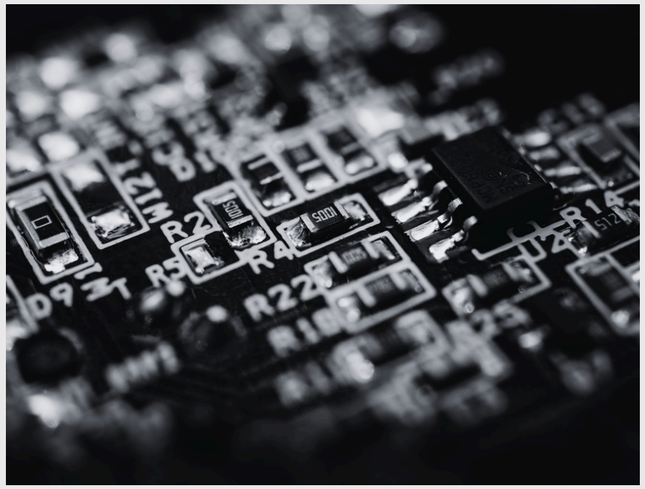
RESULTS

RUNNING



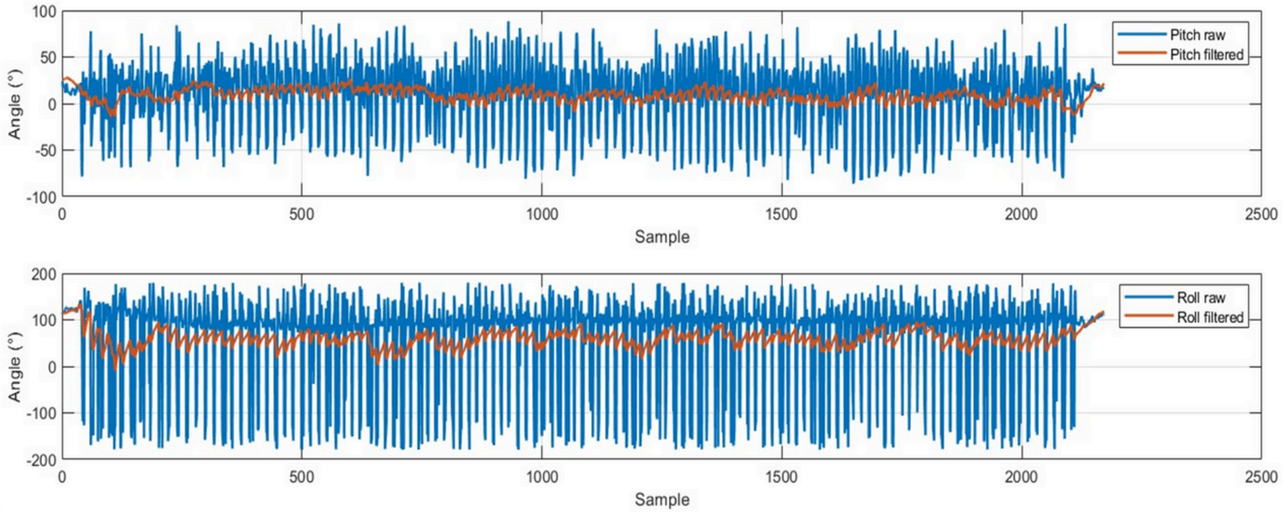


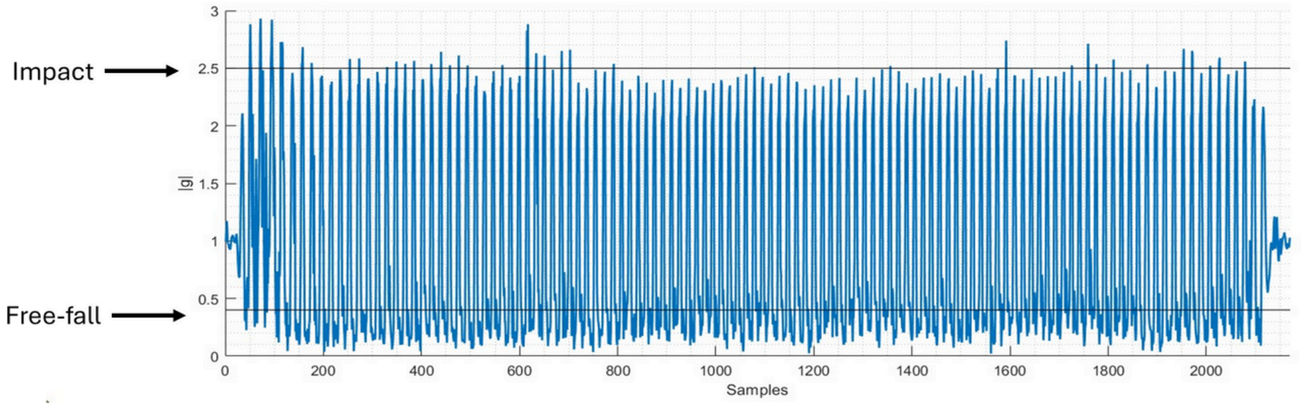




RESULTS

JUMPING





DISCUSSION

STATIC & DYNAMIC TESTS

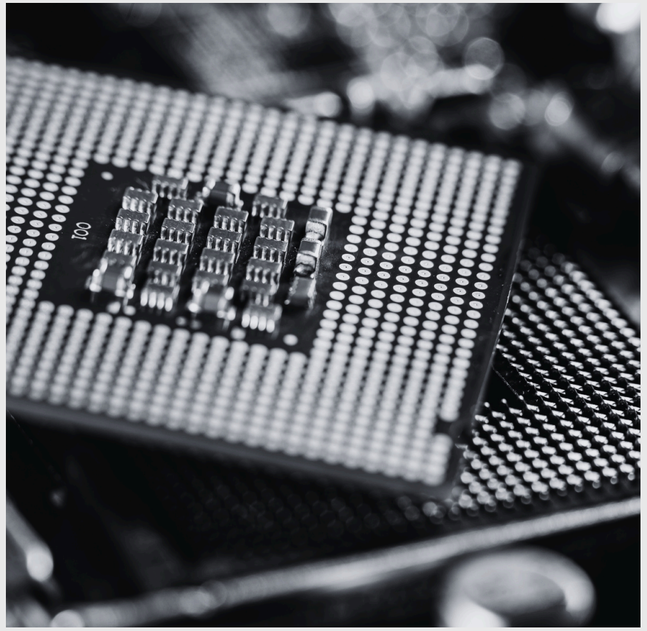
- The falls were properly recognized when |g| and the rotation angles pass the thresholds
- Sitting and lying down change the angle, but |g| is stable
- In low impact tasks, |g| remained within the threshold
- Kalman filter is essential for high impact tasks, avoiding the noisy change of angles being read as a change of position

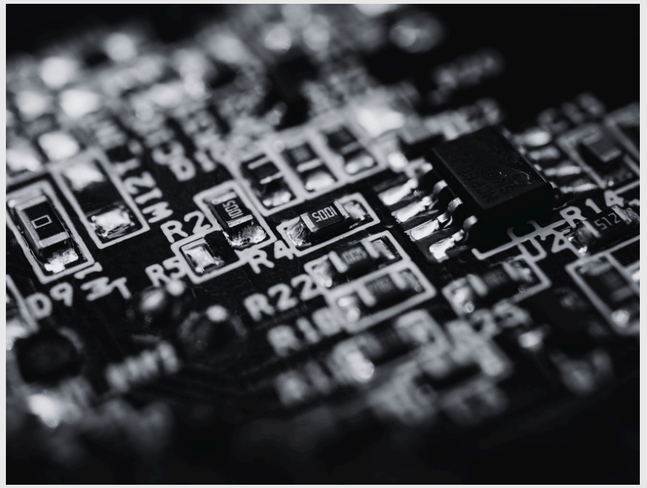
ACCURACY

- From 20 fall experiments, 19 were correctly detected as fall
- Only one was detected as not fall
- 95% of accuracy
- Average of 3.23 seconds from first flag detected to fall detected

COMPARISON WITH OTHER STUDIES

Study	Detection Time (s)	Accuracy
[1]	5	90.83%
[2]	4	98.51%
[3]	3 (threshold part)	95.29%
Our	**3.23**	**95%**





LIMITATIONS & **FUTURE WORK**

LIMITATIONS AND FUTURE IMPROVEMENTS

CURRENT LIMITATIONS

- IMU mounted externally on the enclosure, bringing to possible misalignment and instability.
- Limited number of test scenarios, and few number of experiments
- Detection time influenced by the post-fall observation window.
- A ML model was trained, but not implemented
- No study regarding system autonomy was made

HARDWARE IMPROVEMENTS

- Better PCB and enclosure design with improved stability of the IMU.
- Use a microcontroller with integrated wireless communication.
- Design an efficient power supply circuit.

ALGORITHM IMPROVEMENTS

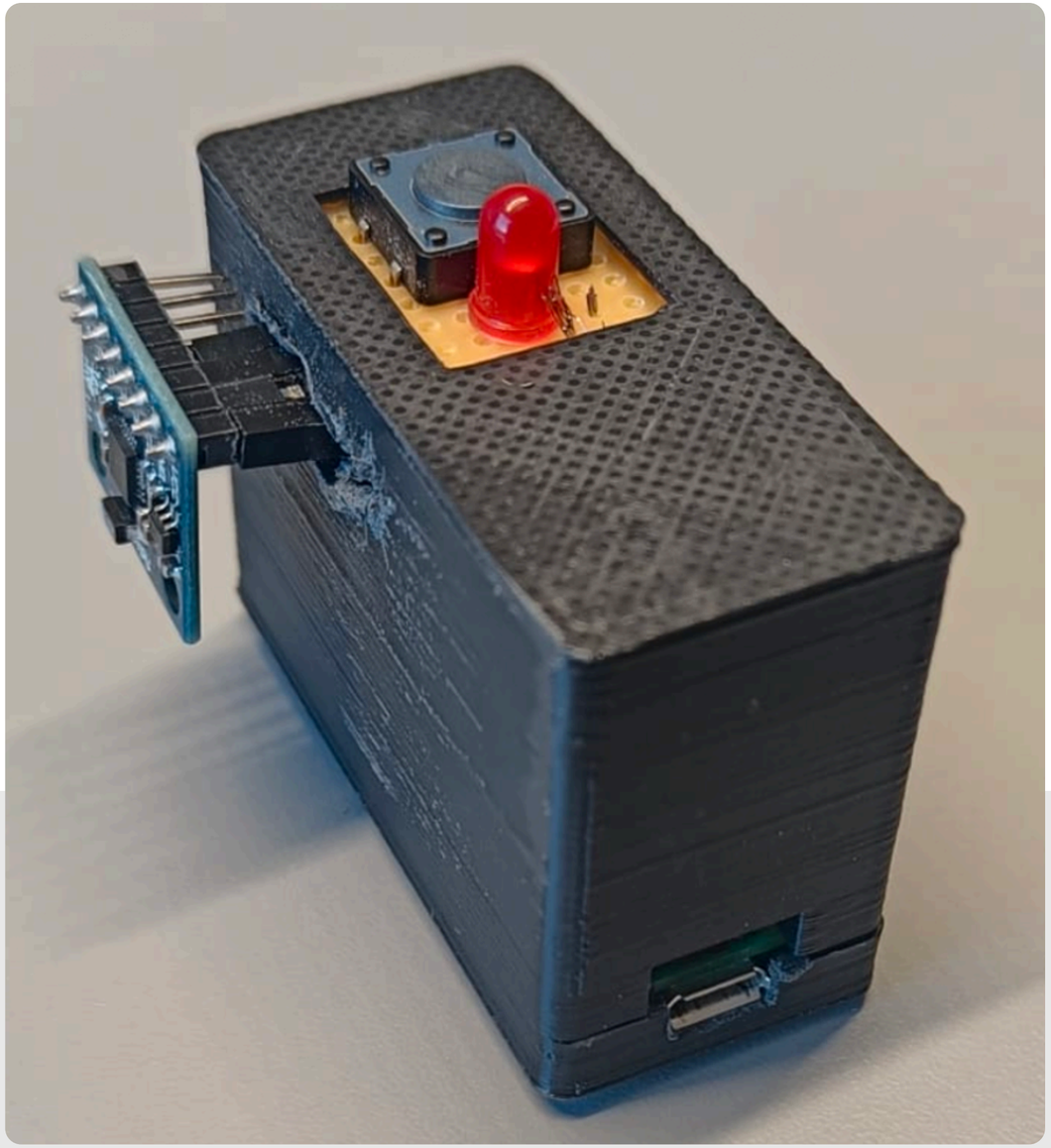
- integration of the trained deep learning model (CNN) on the device for improving the fall detection.
- Reduction on detection time

OTHER IMPROVEMENTS

- Extended testing with more subjects and real life activities (stairs, jogging, jumping)
- Implement the wireless alert system

CONCLUSION

- We developed a working prototype of a wearable fall detection device using minimal hardware.
- The threshold-based algorithm successfully detects falls and rejects the tested non-fall activities.
- The system includes both local feedback (LED + button) and real-time visualization via GUI.
- There is clear potential for further improvements in hardware integration and intelligent detection methods.



THANKS FOR THE ATTENTION

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[2] M. Villa and E. Casilari, “Energy-efficient fall-detection system using lora and hybrid algorithms,” Biomimetics, vol. 10, no. 5, p. 313, 2025.

[3] J. Fernandez-Bermejo, J. Martinez-del Rincon, J. Dorado, X. D. Toro, M. J. Santofimia, and J. C. Lopez, “Edge computing transformers for fall detection in older adults,” International journal of neural systems, vol. 34, no. 05, p. 2450026, 2024.